PICAMERA AND OPENCV

CZ3004/CE3004 Multi-disciplinary Design Project

Dr. Loke Yuan Ren
yrloke@ntu.edu.sg

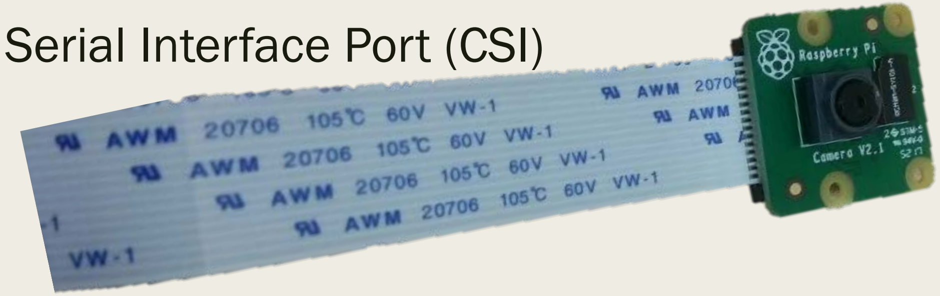
Overview

- Getting Started with Picamera
- Picamera Testing with Python
- OpenCV Installation
- Picamera + Python + OpenCV
- References

Getting Started with PiCamera

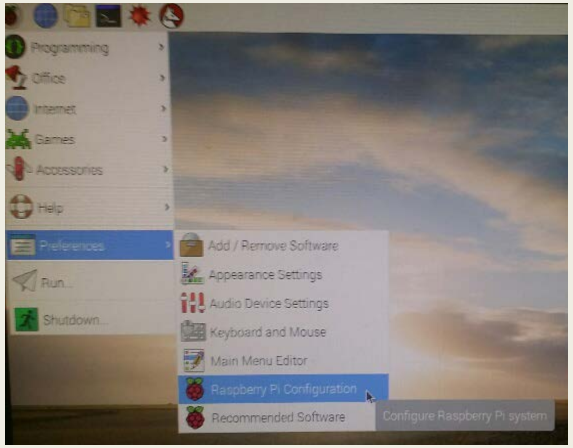
- Step 1: Connect the camera module to your Raspberry Pi

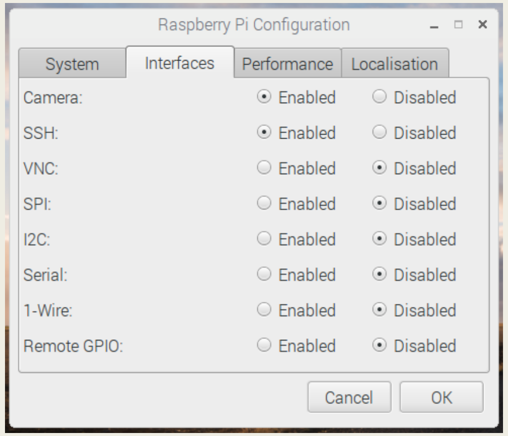
- PiCamera V2.1
  - *8M Pixels fixed focus*
  - *Support 1080p30, 720p60, VGA640x480p90*
  - *Sony IMX219PQ*
  - *Pixel size: 1.12μmx1.12μm*
  - *Optical size: ¼”*
  - *Focus Length: 3.04mm*
  - *HFoV: 62.2^{o} VFoV: 48.8^{o}*
  - *F-Stop: 2.0*
- Interface: Camera Serial Interface Port (CSI)



Getting Started with Picamera

- Step 2: Enable the camera under Rasberry Pi Configuration and reboot your Pi after enable it.



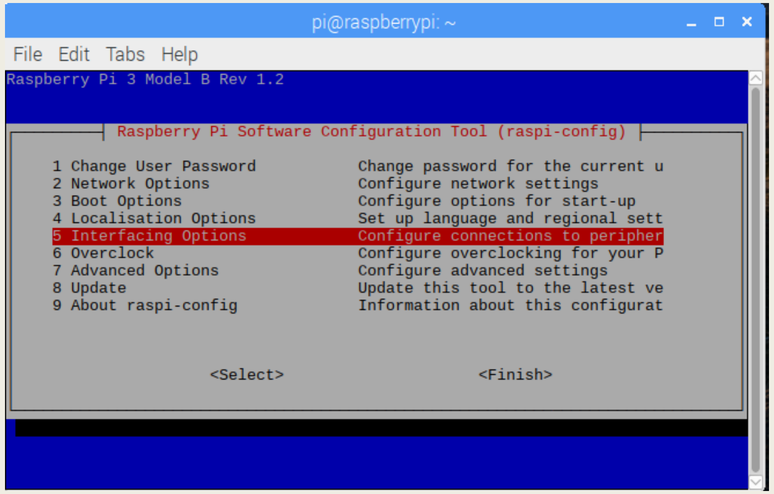


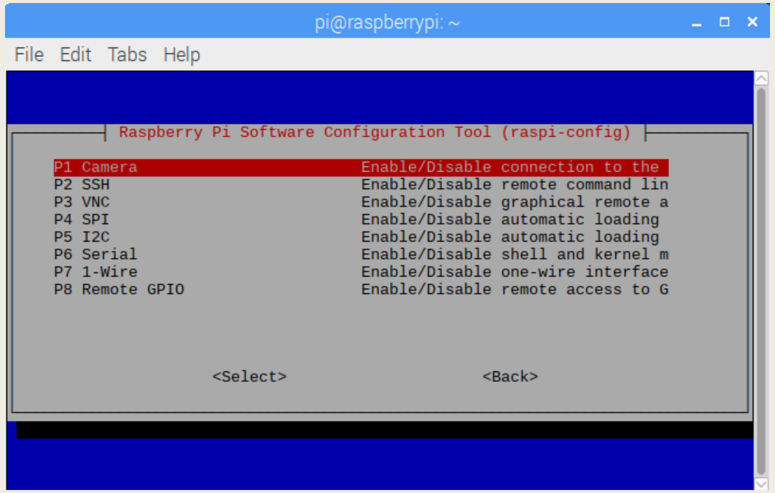
- For CLI user,

```
$ sudo raspi-config
```

- After enable camera,

```
$ sudo reboot
```





Picamera Testing with Python

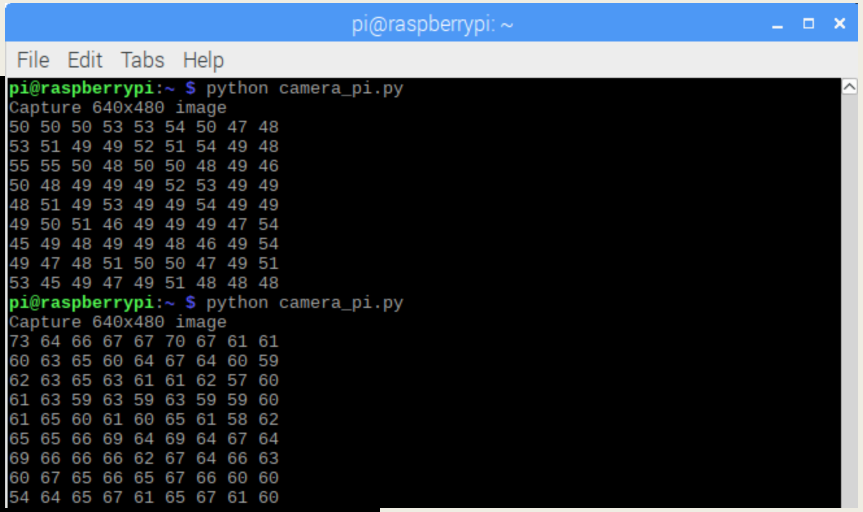
- Step 3: Camera Preview
  - Preview, sleep for 10 seconds, stop preview

```
from picamera import PiCamera
from time import sleep

camera = PiCamera()
camera.start_preview()
sleep(10)
camera.stop_preview()
```

Picamera Testing with Python

- Step 4: Change camera resolution, capture an image and get access to selected pixels. More about PiCamera API: https://picamera.readthedocs.io/en/release-1.13/

```
from picamera import PiCamera
from picamera.array import PiRGBArray

camera = PiCamera()
camera.resolution = (640,480)
output = PiRGBArray(camera)
camera.capture(output,’bgr’)
src = output.array
print(‘Capture %dx%d image’ %(src.shape[1],src.shape[0]))
for i in range(1,10):
    s=''
    for j in range(1,10):
        s=s+repr(src[i,j,1])+' '
    print(s)
```



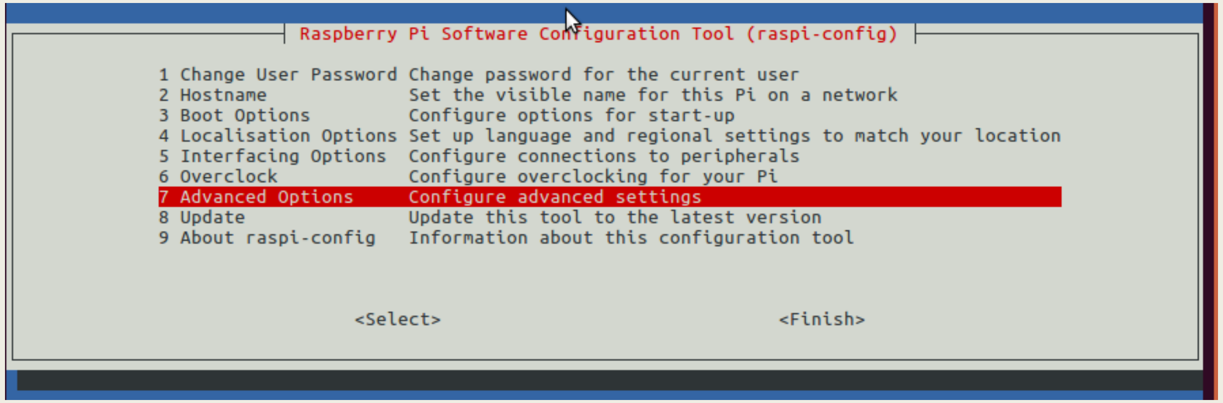
PiCamera

- Now you can preview, capture an image, and storage it
- PiCamera API offers very limited functions for image processing and computer vision
- Next, we will introduce an open-source BSD-licensed library, OpenCV
- OpenCV includes hundreds of computer vision and image processing algorithms

OpenCV Installation

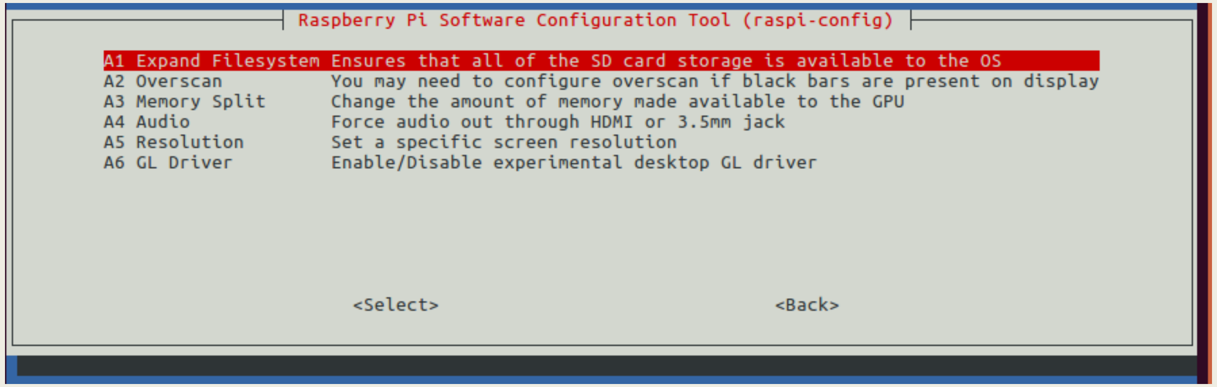
- Step 1: Expand filesystem
- Step 2: Install dependencies
- Step 3: Download the OpenCV source code
- Step 4: Install pip and numpy
- Step 5: Compile and Install OpenCV
- Step 6: Test OpenCV with Python

OpenCV Installation



- Step 1: Expand filesystem

```
$ sudo raspi-config
```



- After that, reboot your Raspberry Pi

```
$ sudo reboot
```

OpenCV Installation

- Step 1: Expand filesystem
  - After reboot the system, please check the disk space

```
$ df -h
```

  - Make sure that you have a few GB available to install OpenCV
  - If you run out of space, you may consider to purge your wolfram-engine and libreoffice. They can return almost 1GB for you.
  - But think twice before you do so

```
$ sudo apt-get purge wolfram-engine
$ sudo apt-get purge libreoffice*
```

OpenCV Installation

- Step 2: Install dependencies
  - *Update and upgrade any existing packages:*

```
$ sudo apt-get update && sudo apt-get upgrade
```

  - *Install developer tools (Installed earlier)*

```
$ sudo apt-get install build-essential pkg-config
```

  - *Install CMake (for installing OpenCV source later)*

```
$ sudo apt-get install cmake
```

OpenCV Installation

- Step 2: Install dependencies
  - *Image I/O packages:*

```
$ sudo apt-get install libjpeg-dev libtiff5-dev libjasper-dev libpng12-dev
```

  - *Video I/O packages:*

```
$ sudo apt-get install libavcodec-dev libavformat-dev libswscale-dev libv4l-dev
$ sudo apt-get install libxvidcore-dev libx264-dev
```

  - *Display I/O packages: for highgui module in OpenCV*

```
$ sudo apt-get install libgtk2.0-dev libgtk-3-dev
```

  - *Linear algebra library and fortran compiler*

```
$ sudo apt-get install libatlas-base-dev gfortran
```

OpenCV Installation

- By default, python2.7 and Python 3 should be installed in your Raspberry Pi. If not,
  - `$ sudo apt-get install python2.7-dev python3-dev`
- Step 3: Download the OpenCV source code
  - *Here we download OpenCV 3.3.0. The latest version is 3.4.x. You can try it.*

```
$ cd ~
$ wget -O opencv.zip https://github.com/Itseez/opencv/archive/3.3.0.zip
$ unzip opencv.zip
$ wget -O opencv_contrib.zip https://github.com/Itseez/opencv_contrib/archive/3.3.0.zip
$ unzip opencv_contrib.zip
```

OpenCV Installation

- Step 4: Install pip and numpy
  - *Before compile the OpenCV source codes, you need to install pip and numpy first*

```
$ wget https://bootstrap.pypa.io/get-pip.py
$ sudo python get-pip.py
$ sudo python3 get-pip.py #for python 3.x user
```

- Here we skip the virtualenv. If you would like to have it, you can refer step 4 and 6 in https://www.pyimagesearch.com/2017/09/04/raspbian-stretch-install-opencv-3-python-on-your-raspberry-pi/

- *It is mainly used for multiple projects in Python 2 and Python 3*
- *Here I assume that you are using Python 2.x only*
- *Install the numpy for numerical processing*

```
$ pip install numpy
```

OpenCV Installation

- Step 5: Compile and Install OpenCV

```
$ cd ~/opencv-3.3.0/
$ mkdir build
$ cd build
$ cmake -D CMAKE_BUILD_TYPE=RELEASE \
    -D CMAKE_INSTALL_PREFIX=/usr/local \
    -D INSTALL_PYTHON_EXAMPLES=ON \
    -D OPENCV_EXTRA_MODULES_PATH=~/opencv_contrib-3.3.0/modules \
    -D BUILD_EXAMPLES=ON ..
```

  - *Configure your swap space size before compiling*

```
$ sudo nano /etc/dphys-swapfile
```

  - *Edit* `CONF_SWAPSIZE=1024`
  - *CTRL+x to save and exit*
  - *Activate the change and compile OpenCV by 4 core (it will take nearly an hour)*

```
$ sudo /etc/init.d/dphys-swapfile stop
$ sudo /etc/init.d/dphys-swapfile start
$ make -j4
```

OpenCV Installation

- Step 5: Compile and Install OpenCV
  - *Install OpenCV*

```
$ sudo make install
$ sudo ldconfig
```

- Step 6: Test OpenCV with Python

```
$ python
>>> import cv2
>>> cv2.__version__
'3.3.0'
>>>
```

  - *Reconfigure your swap space size back to 100*

```
$ sudo nano /etc/dphys-swapfile
```

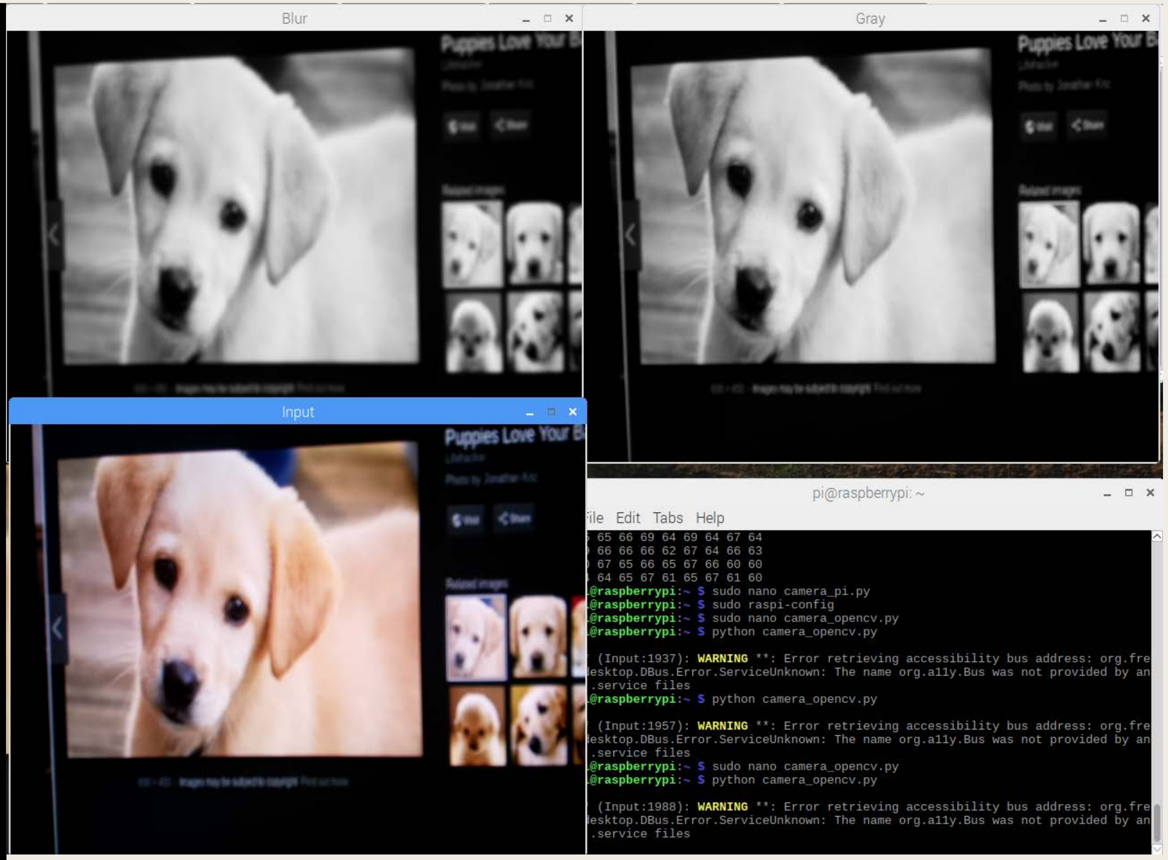
  - *Edit CONF_SWAPSIZE=1024*

```
$ sudo /etc/init.d/dphys-swapfile stop
$ sudo /etc/init.d/dphys-swapfile start
```

Picamera+OpenCV in Python

- Convert color image to gray image. Blur the gray image.
  For more information about OpenCV: https://docs.opencv.org/3.3.0/

```
from picamera import PiCamera
from picamera.array import PiRGBArray
import cv2 as cv
import numpy as np

camera = PiCamera()
camera.resolution = (640,480)

output = PiRGBArray(camera)
camera.capture(output,'bgr')

src = output.array
src_gray = cv.cvtColor(src,cv.COLOR_BGR2GRAY)
dst=cv.blur(src_gray,(5,5))

cv.imshow('Input',src)
cv.imshow('Gray',src_gray)
cv.imshow('Blur',dst)
cv.waitKey(0)
```

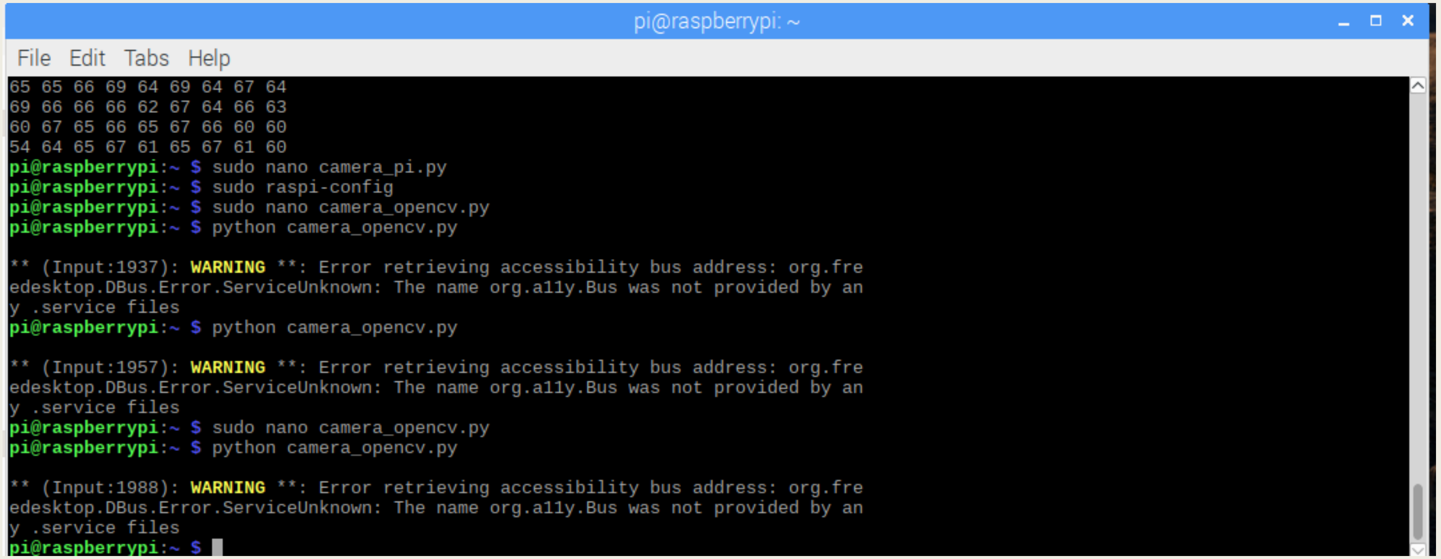


Warning Issue

- If you receive the warning due to using imshow, it is okay.
- You can easily remove it by adding the following line in `~/.bashrc`

```
export NO_AT_BRIDGE = 1
```

- But it is optional.



Reference

- PiCamera: https://projects.raspberrypi.org/en/projects/getting-started-with-picamera
- PiCamera API: https://picamera.readthedocs.io/en/release-1.13/
- OpenCV Installation: https://www.pyimagesearch.com/2017/09/04/raspbian-stretch-install-opencv-3-python-on-your-raspberry-pi/
- OpenCV Documentation: https://docs.opencv.org/3.3.0/